

Experiment ‘p_4_6_find_minimum_seq’ Results

June 5, 2026

Experiment outcome: SUCCESS

Bad responses: 0

Responses containing assume : 0

Responses containing expect : 0

Resolution attempts: 1

Hard fails (resolution): 0

Soft fails (resolution): 0

Verification attempts: 1

Problem Specification

Problem name: p_4_6_find_minimum_seq

Natural language statement: null

Method signature: p_4_6_find_minimum_seq(inputs: seq<int>) returns (min: int)

Ensures

- forall $i :: 0 \leq i < |\text{inputs}| \implies \text{min} \leq \text{inputs}[i]$
- exists $i :: 0 \leq i < |\text{inputs}| \ \&\& \ \text{min} == \text{inputs}[i]$

Requires

- $0 < |\text{inputs}|$

GenAI interactions

Below you will find all interactions between the ‘user’ (program) and the ‘assistant’ (OpenAI).

Program → GenAI

You are given the following task to perform in Dafny:

Write a method to satisfy the below requirements.

The signature should be:

`p_4_6_find_minimum_seq(inputs: seq<int>) returns (min: int)`

The method should respect the following contract:

requires $0 < |inputs|$, ensures forall $i :: 0 \leq i < |inputs| \Rightarrow \text{min} \leq \text{inputs}[i]$,
 ↪ ensures exists $i :: 0 \leq i < |inputs| \ \&\& \ \text{min} == \text{inputs}[i]$

Produce and show only the Dafny body of this method, including the curly braces that
 ↪ surround it. Do not show the signature nor contract. You must not use 'assume' nor
 ↪ 'expect' anywhere in your code.

GenAI → Program

System fingerprint: null

ID: chatcmpl-DnMDuoa62x4jvmrDq3n8GCona37NY

Model: gpt-5.5-2026-04-23

Created at: 1780655870

Finish reason: stop

Usage: 322 tokens in, and 1023 tokens out

```
{
  var idx := 1;
  var minIndex := 0;
  min := inputs[0];

  while idx < |inputs|
    invariant 1 <= idx <= |inputs|
    invariant 0 <= minIndex < idx
    invariant min == inputs[minIndex]
    invariant forall i :: 0 <= i < idx ==> min <= inputs[i]
  {
    if inputs[idx] < min {
      min := inputs[idx];
      minIndex := idx;
    }
    idx := idx + 1;
  }
}
```

Final Program

```
method p_4_6_find_minimum_seq(inputs: seq<int>) returns (min: int)
  requires 0 < |inputs|
  ensures forall i :: 0 <= i < |inputs| ==> min <= inputs[i]
  ensures exists i :: 0 <= i < |inputs| && min == inputs[i]
{
  var idx := 1;
  var minIndex := 0;
  min := inputs[0];

  while idx < |inputs|
    invariant 1 <= idx <= |inputs|
    invariant 0 <= minIndex < idx
    invariant min == inputs[minIndex]
    invariant forall i :: 0 <= i < idx ==> min <= inputs[i]
  {
    if inputs[idx] < min {
      min := inputs[idx];
      minIndex := idx;
    }
    idx := idx + 1;
  }
}
```

Total Token Usage

Input tokens: 322

Output tokens: 1023

Reasoning tokens: 871

Sum of 'total tokens': 1345

Experiment Timings

Overall Experiment started at 1780655869708, ended at 1780655904330, lasting 34622ms (34.62 seconds)

Iteration #1 started at 1780655869710, ended at 1780655904330, lasting 34620ms (34.62 seconds)